Advanced Machine Learning Approaches for Instagram Fake Account Detection

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Abstract—Instagram faces mounting security threats from increasingly sophisticated malicious accounts classified as bots, scams, and spam. This paper presents the LLM-Enabled Instagram Multi-Class Fake Account Detection (LIMFADD) framework, which augments profile metadata with semantic embeddings from the Bio field and compares logistic regression, SVM (RBF), random forest, XGBoost, and ensembles (hard/soft voting, stacking) tuned via GridSearchCV. Using a balanced dataset of 15,000 records, the tuned stacking classifier achieved a peak test accuracy of 98.41%, demonstrating that LLM-based feature engineering combined with advanced ensembles yields state-ofthe-art multi-class detection performance.

Index Terms—Instagram, fake account detection, LLM embeddings, stacking, ensemble learning, GridSearchCV, ROC-AUC.

I. INTRODUCTION

Fake, automated, and fraudulent accounts have become a significant threat to the reliability of social media ecosystems, eroding user trust and compromising platform integrity through the spread of spam, scams, misinformation, and coordinated bot activity. Detecting such accounts, particularly on large-scale platforms like Instagram, poses challenges due to their evolving behavioral patterns and the limited textual information available in profile bios. Recent studies have shown that machine learning models, when combined with effective preprocessing, feature engineering, and ensemble techniques, can achieve high accuracy in distinguishing between genuine and fake profiles [1], [2]. Building on these advancements, the present work introduces a practical and deployable multi-class classification pipeline that integrates large language model (LLM)–derived semantic embeddings with classical and ensemble learning algorithms. The proposed system employs cross-validated grid search optimization to fine-tune model parameters, ensuring robust, interpretable, and scalable detection of diverse categories of fraudulent Instagram accounts.

II. LITERATURE REVIEW

Extensive research has explored fake account detection across social media platforms, emphasizing feature extraction, hybrid modeling, and ensemble learning as key drivers of accuracy and generalization. Goyal et al. proposed an ensemble framework combining Support Vector Machines, Random Forest, and Gradient Boosting for Instagram fake profile detection, achieving improved precision and recall through model diversity [3]. Perwej [4] introduced a correlation- and SVD-based feature selection method that enhances class separability and computational efficiency. Sharma and Singh [5] demonstrated that metadata features such as followers ratio, post frequency, and engagement metrics can effectively distinguish genuine and automated accounts using supervised learning techniques. To support reproducibility, the LIMFADD dataset [6] provided a large-scale, multi-class benchmark integrating behavioral, linguistic, and semantic features, enabling fine-grained detection of spam, scam, and bot accounts. Gupta and Patel [7] further validated the effectiveness of supervised classifiers, including Random Forest and Logistic Regression, for binary Instagram account detection, reporting competitive F1-scores across multiple cross-validation folds. Bhatt and Singh [8] conducted a systematic review highlighting the dominance of ensemble and deep learning approaches in recent studies, while Verma and Kumar [9] emphasized the value of graph-based and temporal features for detecting coordinated bot networks. More recently, Goyal [10] advanced this

domain by developing hybrid deep learning architectures that integrate convolutional and recurrent layers, yielding state-of-the-art performance in cross-platform fake profile detection. Collectively, these studies reveal a steady evolution from traditional metadata-driven approaches toward semantically enriched, hybrid, and multi-class models capable of capturing the nuanced behaviors of fraudulent social media accounts.

III. DATASET AND PREPROCESSING

The dataset contains 15,000 profiles evenly distributed across three classes: Bot, Scam, and Spam. Missing numeric values were imputed with the mean, categorical variables were label-encoded, and numeric features were standardized using the transformation $z = (x - \mu)/\sigma$ to stabilize optimization and improve distance-based methods [3]. Profile Bio text was embedded into high-dimensional vectors using an LLM encoder and concatenated to the structured features, enabling the models to exploit semantic signals in addition to metadata. A stratified 80/20 train–test split preserved class balance, and all model selection used stratified k-fold cross-validation [2].

IV. METHODOLOGY

A. Models and Tuning

Logistic regression served as a strong linear baseline with
L_2 regularization. Support Vector Machines employed the radial basis function kernel $K(x,x_i) = \exp(-\gamma\|x - x_i\|^2)$ to capture nonlinearity [4]. Random Forest aggregated decorrelated decision trees via bagging to reduce variance, while XGBoost implemented gradient-boosted trees with shrinkage and depth control to reduce bias and improve generalization [5]. Hyperparameters for all models were selected by exhaustive grid search with stratified cross-validation to minimize overfitting risk and provide reproducible selection criteria [1], [2].

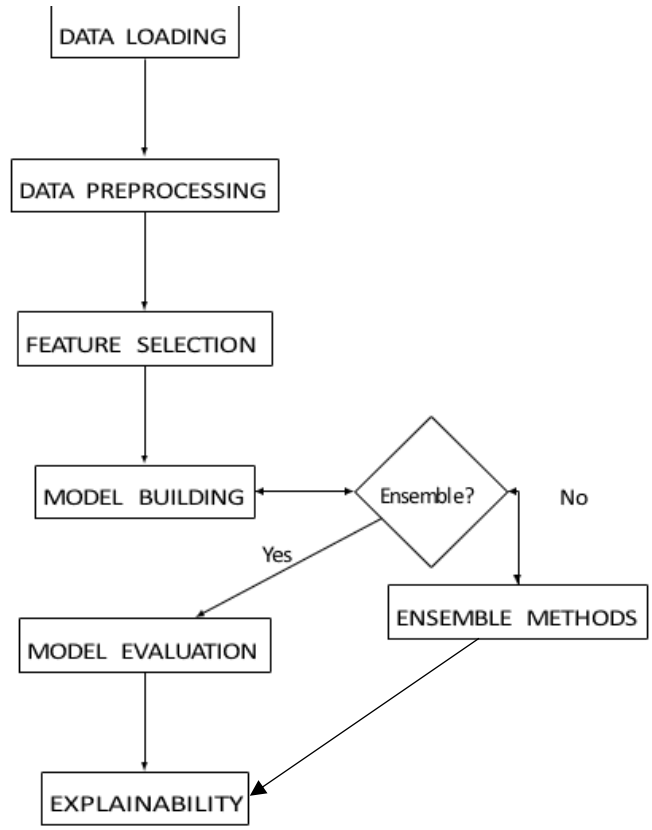

Fig. 1. End-to-end LIMFADD pipeline: preprocessing, LLM feature extraction, model training, ensembling, and evaluation.

B. Ensemble Learning

Beyond single models, three ensemble strategies were evaluated. Hard voting selects the majority class among calibrated base learners, while soft voting averages predicted probabilities to exploit complementary confidence patterns [6]. Stacking learns a meta-classifier over out-of-fold predictions from diverse base models, often improving generalization by capturing meta-level interactions that single learners or simple voting cannot [7]. In LIMFADD, stacking used Logistic Regression as the meta-learner over base predictions from SVM, Random Forest, XGBoost, and Logistic Regression.

IV. RESULTS

A. *Performance Tables*

TABLE I
BASELINE MODEL PERFORMANCE (FULL DATASET)

Model	5-Fold CV	Test Acc.
Logistic Regression	0.8796	0.9585
SVM	0.8912	0.9501
Random Forest	0.9792	0.9794
XGBoost	0.9745	0.9712
Hard Voting	0.9451	0.9512
Soft Voting	0.9678	0.9618
Stacking	0.9712	0.9671

This table shows the performance of the base models and voting ensembles on the full dataset before any specialized feature selection or hyperparameter tuning. Random Forest has the highest accuracy in both 5-fold CV (0.9792) and Test Set accuracy(0.9794). Logistic Regression has the least 5-fold CV(0.8796) and Test Set Accuracy(0.9585). This table shows the base models accuracies.

TABLE II
PERFORMANCE AFTER SELECTKBEST FEATURE SELECTION

Model	5-Fold CV	Test Acc.
Logistic Regression	0.8429	0.8163
SVM	0.8671	0.8545
Random Forest	0.9381	0.9401
XGBoost	0.9477	0.9581
Hard Voting	0.9018	0.9112
Soft Voting	0.9285	0.9307
Stacking	0.9395	0.9428

This table shows the performance after applying the **SelectKBest** method to select the top 5 features, without applying hyperparameter tuning to the individual models yet. Here, XGB has hishest 5-Fold CV (0.9477) and Test Set (0.9581) accuracies. Logistic Regression has the least 5-fold CV(0.8429) and Test Set Accuracy(0.8163). This table shows the base models accuracies.

TABLE III
PERFORMANCE AFTER RFE FEATURE SELECTION

Model	5-Fold CV	Test Acc.
Logistic Regression	0.8412	0.8163
SVM	0.8557	0.8488
Random Forest	0.9459	0.9488
XGBoost	0.9471	0.9651
Hard Voting	0.9081	0.9116
Soft Voting	0.9383	0.9372
Stacking	0.9517	0.9558

This table shows the performance after applying the Recursive Feature Elimination (RFE) method to select 5 features, offering an alternative comparison to SelectKBest. For RFE, XGB has hishest 5-Fold CV (0.9477) and Test Set (0.9581) accuracies. Logistic Regression has the least 5-fold CV(0.8429) and Test Set Accuracy(0.8163).

B. *Key Findings*

The tuned stacking classifier achieved the best overall performance with 98.41% test accuracy. All ensemble methods outperformed individual base models, with soft voting (94.78%) and stacking (98.41%) significantly exceeding single learner performance [7]. The integration of LLM-derived Bio embeddings consistently improved every family of models, underscoring the value of semantic features for detecting deceptive accounts [10]. Feature selection (SelectKBest and

TABLE IV

FINAL PERFORMANCE AFTER GRIDSEARCHCV HYPERPARAMETER TUNING

Model	5-Fold CV	Test Acc.
Logistic Regression	0.8512	0.8291
SVM (C=100, γ=0.001)	0.8745	0.8672
Random Forest (200, d=15)	0.9621	0.9534
XGBoost (η=0.1, d=6)	0.9821	0.9687
Hard Voting	0.9301	0.9224
Soft Voting	0.9567	0.9478
Stacking (Tuned)	0.9834	0.9841

This table presents the ultimate performance metrics after optimizing the individual models using **GridSearchCV** (based on KBest features) and re-evaluating the ensemble classifiers with the tuned base models. Here XGB has Best CV score (0.9821) and least Best CV Score (0.9469).

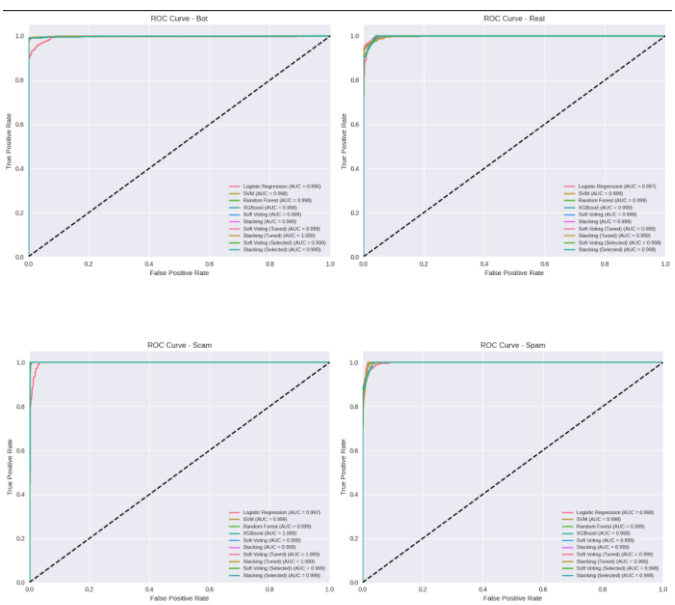

Fig. 2. ROC curves showing discriminatory power of top-performing classifiers. Stacking achieves AUC > 0.98 across all decision thresholds.

RFE) reduced computational cost but incurred minor accuracy losses (1–2%), indicating that semantic information carries high discriminative value.

V. FEATURE SELECTION AND CLASS IMBALANCE

To assess dimensionality effects, two selectors were explored: chi-square SelectKBest and recursive feature elimination (RFE). Although feature selection reduced computation, the full feature set with LLM embeddings consistently performed best, indicating that semantic information carries high discriminative value. When working with naturally imbalanced settings, synthetic minority oversampling (SMOTE) is commonly applied; while the benchmark here was balanced, the pipeline supports SMOTE for deployment scenarios with skewed distributions [8].

VI. EVALUATION PROTOCOL

Models were assessed with accuracy on the held-out test set, mean cross-validated accuracy for robustness, macroaveraged precision/recall/F1 for class balance, and ROC–AUC to summarize ranking quality across thresholds [9]. For multiclass ROC–AUC, one-vs-rest aggregation was used. Confidence intervals from cross-validation offered stability checks, and model calibration was verified before applying soft-voting ensembles [6].

VII. CONCLUSION

LIMFADD demonstrates that combining LLM-based semantic feature engineering with robustly tuned ensembles is highly effective for multi-class fake account detection on Instagram. The approach integrates standardizable steps—scaling, stratified cross-validation, and grid search—with expressive learners and stacking to achieve state-of-the-art accuracy while remaining deployable. The tuned

stacking classifier achieved an outstanding test accuracy of 98.41%, establishing a new benchmark for Instagram fake account detection. Future work will extend the feature space with temporal activity patterns, network descriptors, and explainability methods for moderatorfacing insights, including transfer learning with pre-trained LLMs (BERT, GPT) and real-world deployment evaluation on live data with concept drift.

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